CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 14 : Python Program to Implement Alpha-Beta Pruning

Aim

To develop a Python program that finds the best value of a game tree using the Minimax algorithm with Alpha-Beta pruning.

Algorithm

- Define a recursive function alphabeta(depth, alpha, beta, is_max) and start with alpha = -999 and beta = 999.
- If the maximum depth (3) is reached, return the leaf value.
- At a MAX node, update the value and alpha with the maximum of the child values.
- At a MIN node, update the value and beta with the minimum of the child values.
- If alpha is greater than or equal to beta, prune the remaining branches.
- Call the function from the root as a MAX node and display the best value.

Program

```
def alphabeta(depth, alpha, beta, is_max):
    if depth == 3:
        return depth

    if is_max:
        value = -999
        for i in range(2):
            value = max(value, alphabeta(depth + 1, alpha, beta, False))
            alpha = max(alpha, value)
            if alpha >= beta:
                break
        return value

    else:
        value = 999
        for i in range(2):
            value = min(value, alphabeta(depth + 1, alpha, beta, True))
            beta = min(beta, value)
            if alpha >= beta:
                break
        return value

print("Best value:", alphabeta(0, -999, 999, True))
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/shaik/OneDrive/Documents/14.py ===============
Best value: 3
>>>
```

RESULT

Thus, Alpha-Beta pruning was implemented successfully using Python.